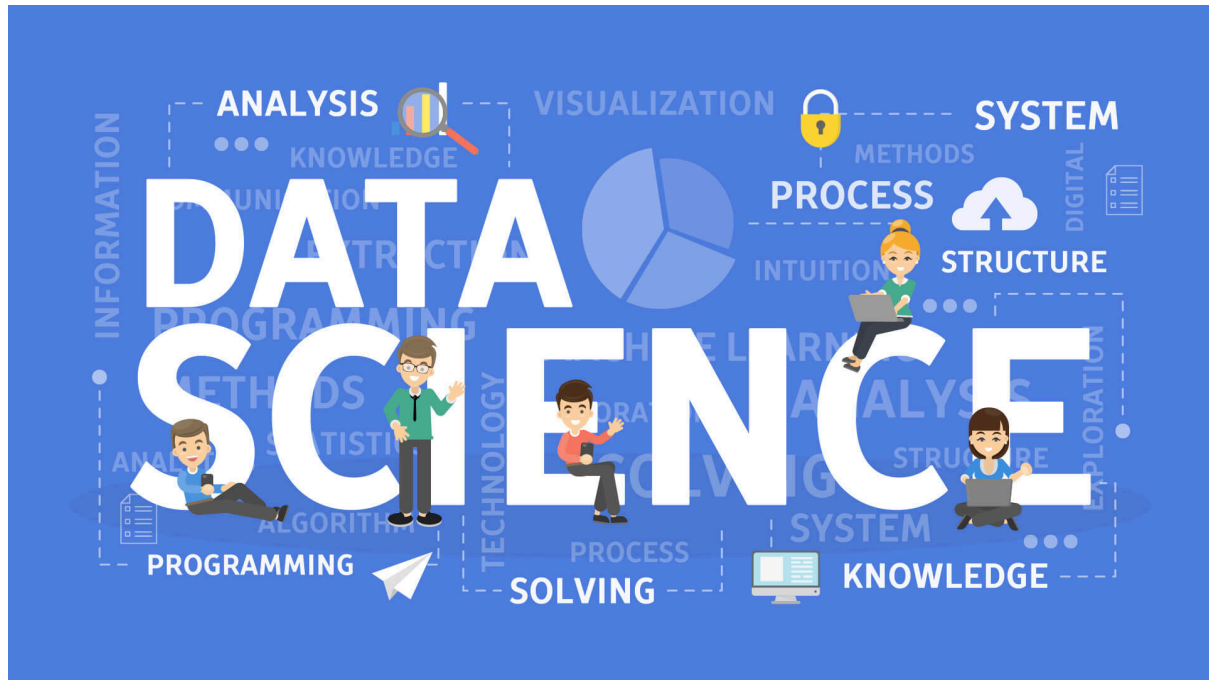


Data Science Job Market Analysis

- Coursework 1 - Data science



Introduction

Data science is one of the fastest-growing fields globally, with organizations increasingly relying on data-driven insights to improve decision-making

In this report, we will discuss the dataset titled "*Data Science Job Postings on Glassdoor*" which has been collected from Kaggle ([Source](#)), published by Rashik Rahman.

This report aims to provide valuable insights for both job seekers and employers. The dataset includes attributes like job titles, salary estimates, company information, and job locations, offering a comprehensive view of the data science job market.

Dataset Description

The dataset contains various job-related attributes, categorized into categorical and numerical data:

- **Categorical Data:** Job Title, Company Name, Location, Industry, Seniority, and other company-related factors like Headquarters, Type of Ownership, and Revenue.
- **Numerical Data:** Salary Estimate (min, max, avg salary), Company Rating (1–5 scale), Company Size.
- **Skills (Binary Features):** Indicates whether specific skills (Python, Excel, etc.) are required (1 = Yes, 0 = No).

Hypothesis & Problem statements

Hypothesis 1: Salary Prediction Based on Job Attributes

Problem Statement:

Salaries in data science roles are impacted by factors such as company rating, job title, and location. Accurately predicting salary ranges allows companies to offer competitive salaries and helps job seekers manage salary expectations.

Business Value:

- **For Employers:** Optimize salary structures to attract talent.
- **For Job Seekers:** Better salary negotiation based on predictive insights.
- **For Economists:** Analyze wage trends by region and industry.

User Story:

As a recruiter, I can adjust salary offers based on company rating and job location to attract top-tier data scientists.

Hypothesis 2: Regional Demand for Data Science Roles

Problem Statement:

The demand for data scientists varies across regions. Identifying areas with the highest demand helps companies focus recruitment efforts and guides job seekers toward cities with the most opportunities.

Business Value:

- For Employers: Focus recruitment efforts on high-demand locations.
- For Job Seekers: Target job searches in cities with the most opportunities.
- For Economists: Inform urban development strategies based on tech hub growth.

User Story:

As an HR leader, identifying emerging tech hubs lets me strategically open offices and prioritize recruitment efforts in cities with the highest demand

Hypothesis 3: Skillset Impact on Seniority and Salary

Problem Statement:

Certain technical skills (e.g., Python) are linked to higher seniority and better salary offers. Understanding these correlations helps job seekers focus on in-demand skills and allows employers to recruit more effectively.

Business Value:

- For Employers: Hire candidates with the right skills for senior roles.
- For Job Seekers: Focus on acquiring skills that lead to better career opportunities.
- For Educational Institutions: Align curriculum with market demand for specific skills.

User Story:

As a program director, I can adjust the curriculum to focus on skills like Python and AWS to ensure graduates are ready for high-salary, senior roles

Faiza Hassan Sheikahmed

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Last Updated: 03.04.25

Data Preprocessing

Step 1: Delete the Index column since it doesn't provide any meaningful information

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Index	Job Title	Salary Estimate	Job Description	Rating	Company Name	Location	Headquarters	Size	Founded	Type of Company	Industry	Sector	Revenue	Competitors	
1	Sr Data Sc	\$137K-\$171K	Description	3.1	Healthfirst	New York, NY	New York, NY	1001 to 5000	1993	Nonprofit	Insurance	Insurance	Unknown	EmblemHealth, UnitedHealth	
2	Data Scie	\$137K-\$171K	Secure ou	4.2	ManTech	Chantilly, VA	Herndon, VA	5001 to 10000	1968	Company	Research	Business Services	\$1 to \$2 b	-1	
3	Data Scie	\$137K-\$171K	Overview	3.8	Analysis Group	Boston, MA	Boston, MA	1001 to 5000	1981	Private Pri	Consulting	Business Services	\$100 to \$1 b	-1	
4	Data Scie	\$137K-\$171K	JOB DESCR	3.5	INFICON	Newton, MA	Bad Ragaz, Switzerland	501 to 1000	2000	Company	Electrical	Manufacturing	\$100 to \$1 b	MKS Instruments, Pfe	
5	Data Scie	\$137K-\$171K	Data Scie	2.9	Affinity Solutions	New York, NY	New York, NY	51 to 200	1998	Company	Advertisin	Business Services	Unknown	Commerce Signals, G	
6	Data Scie	\$137K-\$171K	About Us	4.2	HG Insights	Santa Bar	Santa Bar	51 to 200	2010	Company	Computer	Information Technology	Unknown	-1	
7	Data Scie	\$137K-\$171K	Posting Tit	3.9	Novartis	Cambridge	Basel, Swit	10000+ e	1996	Company	Biotech &	Biotech & Pharmaceuticals	\$10+ billio	-1	
8	Data Scie	\$137K-\$171K	Introducti	3.5	iRobot	Bedford, MA	Bedford, MA	1001 to 5000	1990	Company	Consumer	Retail	\$1 to \$2 b	-1	
9	Staff Data	\$137K-\$171K	Intuit is se	4.4	Intuit - Data	San Diego, CA	Mountain View, CA	5001 to 10000	1983	Company	Computer	Information Technology	\$2 to \$5 b	Square, PayPal, H&R	
10	Data Scie	\$137K-\$171K	Ready to v	3.6	XSELL Technologies	Chicago, IL	Chicago, IL	51 to 200	2014	Company	Enterprise	Information Technology	Unknown	-1	
11	Data Scie	\$137K-\$171K	Join our te	4.5	Novetta	Herndon, VA	Mc Lean, VA	501 to 1000	2012	Company	Enterprise	Information Technology	\$100 to \$1 b	Leidos, CACI Internat	
12	Data Scie	\$137K-\$171K	About Us	4.7	1904labs	Saint Louis, MO	Saint Louis, MO	51 to 200	2016	Company	IT Services	Information Technology	Unknown	Slalom, Daugherty Bu	
13	Data Scie	\$137K-\$171K	*Organiza	3.7	PNNL	Richland, WA	Richland, WA	501 to 5000	1965	Government	Energy	Oil, Gas, Energy & Utilities	\$500 milli	Oak Ridge National Li	
14	Data Mod	\$137K-\$171K	POSITION	3.1	Old World Industries	Northbrook, IL	Northbrook, IL	201 to 500	1973	Company	Chemical	Manufacturing	\$1 to \$2 b	-1	
15	Data Scie	\$137K-\$171K	Position D	3.4	Mathematica Policy Res	Washington, DC	Princeton, NJ	1001 to 5000	1986	Company	Consulting	Business Services	\$100 to \$1 b	-1	
16	Data Scie	\$137K-\$171K	*****P	4.4	Guzman & Griffin Techn	Washington, DC	Mays Lane, VA	1 to 50 e	1997	Company	Federal Ag	Government	Unknown	-1	
17	Data Scie	\$137K-\$171K	We are an	4.1	Upside Business Travel	Remote	Washington, DC	51 to 200	2015	Company	Internet	Information Technology	Unknown	-1	
18	Data Scie	\$137K-\$171K	Job Succes	3.5	Buckman	Memphis, TN	Memphis, TN	1001 to 5000	1945	Company	Chemical	Manufacturing	\$500 milli	-1	
19	Data Anal	\$137K-\$171K	The Data	4.2	Insight Enterprises, Inc.	Piano, TX	Tempe, AZ	5001 to 10000	1988	Company	Enterprise	Information Technology	\$5 to \$10 b	CDW, PCM, SHI Inter	
20	Medical La	\$137K-\$171K	Responsib	3.5	Tower Health	West Gro	Reading, PA	5001 to 10000	2017	Nonprofit	Health Cai	Health Care	Unknown	-1	
21	Data Scie	\$137K-\$171K	Role Descr	3.2	Triplebyte	New York, NY	San Francisco, CA	51 to 200	2015	Company	Computer	Information Technology	Unknown	-1	
22	Data Scie	\$137K-\$171K	PulsePoin	4.3	PulsePoint	New York, NY	New York, NY	51 to 200	2011	Company	Internet	Information Technology	\$100 to \$1 b	Crossix Solutions Inc.	
23	Human Fa	\$137K-\$171K	Human Fa	3.5	Exponent	Phoenix, AZ	Menlo Park, CA	1001 to 5000	1967	Company	Consulting	Business Services	\$100 to \$1 b	-1	
24	Business In	\$137K-\$171K	Position	3.5	Guardian Life	Appleton, WI	New York, NY	5001 to 10000	1860	Company	Insurance	Insurance	\$5 to \$10 b	Northwestern Mutua	
25	Data Scie	\$137K-\$171K	What's sig	3.4	Spectrum Communicati	Chicago, IL	Chicago, IL	51 to 200	1992	Company	Advertisin	Business Services	\$10 to \$2 b	-1	
26	Data Scie	\$137K-\$171K	Job Descr	4.7	Oversight Systems	Atlanta, GA	Atlanta, GA	51 to 200	2003	Company	Computer	Information Technology	\$25 to \$50 b	-1	

Step 2: I cleaned and transformed the Salary Estimate column by extracting the minimum, maximum, and average salaries into separate numerical columns for easier analysis and comparison.

C2

:

X

✓

fx

=SUBSTITUTE(LEFT(B2, FIND("-", B2)-1), "K", "000")*1

	A	B	C	D	E	F	G	H	I
1	Job Title	Salary Estimate	min_sala	max_sala	avg_sala	Job Desc	Rating	Compan	Location
2	Sr Data Scie	137K-171K	137000	171000	154000	Description	3.1	Healthfirst	New York
4	Data Scientist	137K-171K	137000	171000	154000	OverviewAr	3.8	Analysis Gr	Boston, I
5	Data Scientist	137K-171K	137000	171000	154000	JOB DESCR	3.5	INFICON3	Newton,
8	Data Scientist	137K-171K	137000	171000	154000	Posting Title	3.9	Novartis	Cambrid
9	Data Scientist	137K-171K	137000	171000	154000	Introduction	3.5	iRobot3	Bedford,
11	Data Scientist	137K-171K	137000	171000	154000	Ready to wr	3.6	XSELL Tech	Chicago,
14	Data Scientist	137K-171K	137000	171000	154000	*Organizati	3.7	PNNL3	Richland
15	Data Model	137K-171K	137000	171000	154000	POSITION F	3.1	Old World I	Northbro
16	Data Scientist	137K-171K	137000	171000	154000	Position De	3.4	Mathematic	Washing
19	Data Scientist	137K-171K	137000	171000	154000	Job Succes	3.5	Buckman3	Memphis
21	Medical Lab	137K-171K	137000	171000	154000	Responsibil	3.5	Tower Healt	West Gro
22	Data Scientist	137K-171K	137000	171000	154000	Role Descri	3.2	Triplebyte	New York

Step 3: I categorized job titles by seniority, identifying roles containing 'Senior' or 'Sr' and labeling them accordingly for clearer analysis

B2												:	✕ ✓ fx	=IF(OR(ISNUMBER(SEARCH("Senior", A2)), ISNUMBER(SEARCH("Sr", A2))), "Senior", "NA")											
	A	B	C	D	E	F	G	H	I	J	K	L													
1	Job Title	Seniority	Salary Estima	min_sala	max_sala	avg_sala	Job Desc	Rating	Compan	Location	Headqua	Size													
2	Sr Data Scientist	Senior	137K-171K	137000	171000	154000	Description	3.1	Healthfirst	New York, NY	New York, NY	1001 to 5000													
4	Data Scientist NA		137K-171K	137000	171000	154000	OverviewAr	3.8	Analysis Gr	Boston, MA	Boston, MA	1001 to 5000													
5	Data Scientist NA		137K-171K	137000	171000	154000	JOB DESCR	3.5	INFICON3	Newton, MA	Bad Ragaz, Switzerland	501 to 1000													
8	Data Scientist NA		137K-171K	137000	171000	154000	Posting Title	3.9	Novartis	Cambridge, MA	Basel, Switzerland	10000+ employees													
9	Data Scientist NA		137K-171K	137000	171000	154000	Introduction	3.5	iRobot3.5	Bedford, MA	Bedford, MA	1001 to 5000													
14	Data Scientist NA		137K-171K	137000	171000	154000	Ready to wr	3.6	XSELL Tech	Chicago, IL	Chicago, IL	51 to 200													

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Step 4: I decided to delete the founded, Competitor, Sector and headquarter column since I don't need it for my analysis.

Step 5: I created binary columns in Excel to identify the presence or absence of specific skills e.g "Python," in job descriptions.

=IF(ISNUMBER(SEARCH("Python", G2)), 1, 0)								
P	Q	R	S	T	U	V	W	
	python	excel	hadoop	spark	aws	tableau	big_data	
	0	0	0	0	1	0	0	
ion(USD)	0	0	1	0	0	0	1	
million(USD)	1	1	0	0	1	0	0	
million(USD)	1	1	0	0	1	0	0	
	1	1	0	0	0	0	0	
	1	1	1	1	0	0	0	
n(USD)	1	0	0	0	0	0	0	

Step 6: Cleaned the Company Name column by removing ratings and replaced missing values with "NA" throughout the dataset.

=IFERROR(AVERAGE(O2, E2), "")																										
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U						
JobTitle	Seniority	Salary	min_salary	max_salary	avg_salary	JobDescription	Rating	Company Name	Location	Size of employees	Type of ownership	Industry	Revenue	python	excel	hadoop	spark	aws	tableau	big data						
1 DataScientist	Senior	137-171	137000	171000	154000	Security/infrastructure	3.1	HealthFirst	New York, NY	1001-5000	Nonprofit Organization	Insurance/Car	NA	0	0	0	0	1	0	0						
1 DataScientist	NA	137-171	137000	171000	154000	Security/infrastructure	4.2	MarTech	Chenilly, VA	5001-10000	Company - Public	Research&Devel	1-2billion(USD)	0	0	1	0	0	0	0	1					
1 DataScientist	NA	137-171	137000	171000	154000	Overview/analysis	3.8	AnalysisGroup	Boston, MA	1001-5000	Private Practice / Fin	Consulting	100-500million(	1	1	0	0	0	1	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	JOBDESCRIPTION	3.5	IN FICON	Newton, MA	501-1000	Company - Public	Electrical&Ele	100-500million(	1	1	0	0	0	1	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	Outsourcing/infrastructure	2.9	AffinityHealth	New York, NY	51-200	Company - Private	Advertising&M	NA	1	1	0	0	0	0	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	AboutUs/Headquarter	4.2	McKinsey	Santa Barbara, CA	51-200	Company - Private	Computer&Man	NA	1	1	1	1	1	1	0	0					
1 DataScientist/Mac	NA	137-171	137000	171000	154000	PostgreSQL/DataSci	3.9	Novartis	Cambridge, MA	10000+	Company - Public	Biotech&Phar	10+million(USD)	1	0	0	0	0	0	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	Introduction/entry	3.5	Roche	Bellford, NJ	1001-5000	Company - Public	Consumer&E	1-2billion(USD)	1	0	0	0	0	0	0	0					
0 StaffDataScientist	NA	137-171	137000	171000	154000	Infrastructure/infrastructure	4.4	Health Data	San Diego, CA	5001-10000	Company - Public	Computer&Man	2-5billion(USD)	0	0	0	0	0	0	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	Ready-writes/infrastructure	3.6	XSELL Technologies	Chicago, IL	51-200	Company - Private	EnterpriseSoft	NA	1	0	0	0	0	0	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	Join our team/infrastructure	4.5	Novetta	Hemeton, VA	501-1000	Company - Private	EnterpriseSoft	100-500million(	1	1	0	0	1	0	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	AboutUs/infrastructure	4.7	1004labs	St. Louis, MO	51-200	Company - Private	IT Services	NA	1	0	0	1	0	0	0	0					
1 DataScientist/Staff	NA	137-171	137000	171000	154000	*Organization/infrastructure	3.7	PNVIL	Richland, WA	1001-5000	Government	Energy	500million-1billion(	0	1	0	0	0	0	0	0					
1 DataModeler	NA	137-171	137000	171000	154000	POSITION/PURPOSE	3.1	Oldworldindustries	Northbrook, IL	201-500	Company - Private	Chemical&Man	1-2billion(USD)	1	1	0	0	0	1	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	PositionDescription	3.4	MathematicaPolicy&Research	Washington, DC	1001-5000	Company - Private	Consulting	100-500million(	1	1	0	0	1	0	0	0					
7 ExperiencedData	NA	137-171	137000	171000	154000	*****PlaceAs	4.4	QuintanaRoo-InfTech	Washington, DC	NA	Company - Private	FederalAgenc	NA	1	1	1	1	1	0	1	0					
1 DataScientist-Cor	NA	137-171	137000	171000	154000	Weareanambitious	4.1	UpserveBusinessTravel	Remote	51-200	Company - Private	Internet	NA	1	0	0	0	0	0	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	JoinSuccessProfile	3.5	Buckman	Memphis, TN	1001-5000	Company - Public	Chemical&Man	500million-1billion(	1	0	1	0	0	0	0	0					
0 DataAnalyst	NA	137-171	137000	171000	154000	TheDataAnalytics	4.2	Insight&Enterprises, Inc	Plano, TX	5001-10000	Nonprofit Organization	EnterpriseSoft	5-10billion(USD)	0	1	0	0	0	0	0	0					
1 MedicalDataScientist	NA	137-171	137000	171000	154000	Responsibilities/infrastructure	3.5	TowerHealth	West Grove, PA	5001-10000	Company - Private	Health&Care	NA	0	0	0	0	0	0	0	0					
2 DataScientist	NA	137-171	137000	171000	154000	RoleDescription/infrastructure	3.2	Truist	New York, NY	51-200	Company - Private	Computer&Man	NA	0	0	0	0	0	0	0	0					
1 DataScientist/Mac	NA	137-171	137000	171000	154000	PulsePoints/infrastructure	4.3	PulsePoint	New York, NY	51-200	Company - Public	Internet	100-500million(	1	0	1	1	0	0	0	1					
1 HumanResources	NA	137-171	137000	171000	154000	HumanResources/infrastructure	3.5	Exponent	Phoenix, AZ	1001-5000	Company - Private	Consulting	100-500million(	0	1	0	0	0	0	0	0					
5 BusinessIntelligence	NA	137-171	137000	171000	154000	PositionSummary/infrastructure	3.5	GuardianLife	Appleton, WI	5001-10000	Company - Private	Insurance&Car	1-10billion(USD)	0	0	0	0	0	0	0	0					
1 DataScientist	NA	137-171	137000	171000	154000	WhatE'ssignificant	3.4	SpectraCommunications	Chicago, IL	51-200	Company - Private	Advertising&M	10-25million(USD)	0	0	0	0	0	0	0	0					
7 DataScientist	NA	137-171	137000	171000	154000	JobDescription/infrastructure	4.7	OversightSystems	Atlanta, GA	51-200	Company - Private	Computer&Man	25-50million(USD)	1	1	0	0	0	0	0	0					
8 DataScientist-R	NA	137-171	137000	171000	154000	JustifiedDataScientist	4.2	USQ	Orlando, FL	51-200	Nonprofit Organization	Investment&F	100-500million(	1	1	1	1	1	1	1	1					
9 DataScientist-H	NA	137-171	137000	171000	154000	Lexington, MA, USA/infrastructure	3.8	MIT Lincoln Laboratory	Lexington, MA	5001-10000	Company - Private	Aerospace&Def	NA	1	0	1	1	1	0	0	1					
0 DataScientist	NA	137-171	137000	171000	154000	ingfisherSystems, IL	4.5	ingfisherSystems	McLean, VA	51-200	Company - Private	FederalAgenc	25-50million(USD)	1	0	0	0	0	0	1	0					
1 DataScientist	NA	137-171	137000	171000	154000	Formation/infrastructure	2.8	Formation	San Francisco, CA	51-200	Private Practice / Fin	EnterpriseSoft	NA	0	1	0	1	0	0	1	0					
2 DataScientist	NA	75-131	75000	131000	103000	Overview/analysis	3.8	AnalysisGroup	Boston, MA	1001-5000	Company - Public	Consulting	100-500million(	1	0	0	0	0	1	0	0					
3 DataScientist/Mac	NA	75-131	75000	131000	103000	PostgreSQL/DataSci	3.9	Novartis	Cambridge, MA	10000+	Company - Private	Biotech&Phar	10+million(USD)	1	0	0	0	0	0	0	0					
1 DataScientist	NA	75-131	75000	131000	103000	ConnerHealth/infrastructure	5	ConnerHealth	Boston, MA	NA	Company - Private	IT Services	NA	1	1	0	0	0	0	0	0					
5 SeniorResearcher	Senior	75-131	75000	131000	103000	Acetylsalicylic acid	4.8	Acetylsalicylic acid	Sheboygan, WI	1001-5000	Company - Private	Insurance&Car	1-2billion(USD)	0	0	0	0	0	0	0	0					
6 DataEngineer	NA	75-131	75000	131000	103000	ChiefSoftwareEngine	3.6	Chief	United States	201-500	Company - Private	EnterpriseSoft	50-100million(USD)	0	1	0	0	0	1	0	1					
1 AssociateDataSci	NA	75-131	75000	131000	103000	PugetSoundEnergy	3.3	PugetSoundEnergy	Bothell, WA	1001-5000	Company - Private	Utilities	2-5billion(USD)	0	1	1	1	1	1	0	0					
8 BusinessIntelligence	NA	75-131	75000	131000	103000	TheBusinessIntell	2.7	SanctiBiosGlobal	Lincoln, NE	1001-5000	Company - Private	EnterpriseSoft	50-100million(USD)	1	1	0	0	0	0	1	0					

Proposed Solution-

For my proposed solution, I have chosen the K-Nearest Neighbors (KNN) algorithm, a supervised learning technique. KNN is an ideal method for this task because it can handle both regression and classification problems, which aligns well with the varied nature of the hypotheses.

The first hypothesis focuses on salary prediction. Here, the goal is to predict salary ranges based on job-related attributes, such as company rating, job title, location, and required skills. KNN can address this regression problem by identifying jobs with similar features and providing a predicted salary range based on the salary of the nearest neighbors. If we input job attributes, the algorithm will calculate the nearest neighbors and output the most likely salary estimate.

The second hypothesis addresses regional demand for data science roles. In this case, the objective is to classify regions based on the demand for data scientists, using factors such as company size, industry, revenue, and location. KNN can perform this classification by comparing a given location's features to those of similar regions and predicting whether the demand is high, medium, or low. By identifying patterns in similar regions, KNN can effectively predict the demand for data scientists in various locations.

The third hypothesis involves the impact of skill sets on seniority and salary. The hypothesis examines how technical skills (e.g., Python, AWS, Hadoop) influence job seniority and salary levels. KNN will help classify jobs into seniority levels (e.g., junior, senior) and predict salary ranges based on the presence of specific skills. For example, jobs requiring advanced skills like Python or excel will be classified as having higher seniority and salary, based on the nearest neighbors with similar skill requirements.

KNN is particularly suited for this task due to its ability to handle both numerical and categorical data. The dataset includes attributes such as salary estimates (numerical) and job titles or locations (categorical), which can be preprocessed appropriately for KNN. The simplicity and efficiency of KNN, combined with its ability to adapt to multiple types of problems, make it the most appropriate technique to analyze the job-related data in this task.

In conclusion, KNN is a versatile and effective method for predicting salaries, analyzing regional demand, and evaluating the impact of skill sets on job attributes, making it the ideal choice for this project.